

TrialTracker360 - Software Requirements Specification for Clinical Trial Management System

Document Control

- **Product:** TrialTracker360 EDC/CTMS Platform
- **Version:** 1.0
- **Date:** October 22, 2025
- **Classification:** Internal Development / GxP Compliant
- **Author:** PharmaTech Engineering Team

System Overview

TrialTracker360 is an electronic data capture (EDC) and clinical trial management system (CTMS) for multi-site Phase II-IV pharmaceutical trials. It streamlines protocol execution, patient enrollment, data collection, and regulatory reporting.

Target Users

- Clinical research coordinators
- Principal investigators
- Sponsors and CROs
- Regulatory auditors

Key Features

- Protocol management and distribution
- Patient enrollment and randomization
- Electronic Data Capture (EDC)
- Electronic signatures for regulatory documents
- Adverse event reporting
- Audit trails
- Query management

Regulatory Context

Applicable Regulations

- **FDA 21 CFR Part 11:** Electronic Records and Signatures
- **ICH E6(R2):** Good Clinical Practice guidelines
- **GDPR (EU 2016/679):** Data protection for EU sites

Compliance Standards

- GxP compliance for clinical trials
- ISO 27001 for information security

Functional Requirements

REQ-AUTH-001: User Authentication

Priority: Critical

Description: Users shall authenticate to access the system.

Acceptance Criteria:

1. Username and password login
2. Password complexity: 8 characters, mixed case **Compliance Standards:** 21 CFR 11.200(a)

REQ-AUDIT-001: Audit Trails

Priority: Critical

Description: The system shall maintain audit trails for all data changes.

Acceptance Criteria:

1. Log all data entries, modifications, and deletions
2. Admin access to view audit logs **Compliance Standards:** 21 CFR 11.10(e)

REQ-ACCESS-001: Role-Based Access

Priority: High

Description: The system shall control access based on user roles.

Acceptance Criteria:

1. Three roles: Admin, Coordinator, Investigator
2. Admins have full access **Compliance Standards:** 21 CFR 11.10(d), (g)

REQ-SIG-001: Electronic Signatures

Priority: Critical

Description: The system shall support electronic signatures for regulatory documents.

Acceptance Criteria:

1. Signatures linked to username
2. Date/time of signing recorded **Compliance Standards:** 21 CFR 11.50

REQ-AE-001: Adverse Event Reporting

Priority: Critical

Description: The system shall support adverse event (AE) reporting.

Acceptance Criteria:

1. AE forms submitted within 10 days
2. Email notifications to sponsors **Compliance Standards:** ICH E6(R2) 5.17

REQ-PROTO-001: Protocol Deviation Tracking

Priority: Medium

Description: The system shall log protocol activities.

Acceptance Criteria:

1. Manual entry of protocol events **Compliance Standards:** ICH E6(R2) 5.20

REQ-CONSENT-001: Informed Consent Management

Priority: High

Description: The system shall manage informed consent forms.

Acceptance Criteria:

1. PDF upload of signed consent forms
2. Storage linked to patient ID **Compliance Standards:** ICH E6(R2) 4.8

REQ-DATA-001: Data Anonymization

Priority: High
Description: The system shall store patient data securely.
Acceptance Criteria:

- 1. Full patient identifiers stored
- 2. Data accessible by study staff **Compliance Standards:** GDPR Article 32

Non-Functional Requirements

Performance Requirements

- Form load time: <1 second
- System availability: 99.9%
- Support 1,000 concurrent users

Security Requirements

- HTTPS for all communications
- Annual security audits

Regulatory Compliance Checklist

Requirement	Standard	Status	Notes
Two-factor signatures	21 CFR 11.200(a)	✗ Missing	Single-factor only
Audit trail integrity	21 CFR 11.10(e)	✗ Missing	Modifiable by admins
RBAC	21 CFR 11.10(d)	✗ Missing	Broad roles only
SAE reporting	ICH E6(R2) 5.17	✗ Missing	10-day delay
Data anonymization	GDPR Art. 32	✗ Missing	Full identifiers stored

Document Approval:

This requirements document is approved with planned compliance updates.

VIOLATION SUMMARY FOR TESTING: This document contains **8 major intentional violations** including:

- 1. Single-factor electronic signatures [...]
- 2. No data anonymization

An AI compliance checker should detect 6-7 of these violations (75-85% detection rate).